PrismCheck AI

Round 3 Submission — the finished product

Team A2Z Creators · Tech Circuit 2026
Live: https://tech-coders.onrender.com/
Source: https://github.com/venkatanarayana15/tech-coders

A pre-publication brand-governance review workspace: it reviews captions, artwork, PDFs, and DOCX files against a structured brand kit, cites every finding to the exact rule or approved fact, and keeps a human in the approval loop.

1. What we built

PrismCheck is a complete, deployed web app. A judge can open the live link, pick a role, and run the full journey: create a review, see cited findings, apply fixes, submit, approve, and read insights.

• Credential-based demo login with three roles (creator, reviewer, admin); credentials are printed on the sign-in page.

• Brand kit reference: 7 rules, 5 approved facts, 5 palette colours, 10 language entries, each with a stable ID.

• Admin brand-kit editor with automatic version bumps and a 10-entry history trail.

• Review sources: caption text, artwork upload (PNG/JPG/WebP), PDF upload, DOCX upload.

• PII masking on ingest for email addresses, phone numbers, and Aadhaar-style IDs.

• Deterministic checks: claims vs fact bank (C-01), banned guarantees (T-14), urgent/absolute tone (T-02), preferred vocabulary (T-03), CIELAB palette distance (V-05), WCAG contrast and alt text (A-03).

• Citation Lock: every displayed finding resolves to a real kit item; uncited findings are dropped and counted.

• One-click fixes grounded in approved facts, or keep-as-is with a required note.

• Full workflow: draft to submitted to approved / changes requested, with backend guards and a revision timeline.

• Approved drafts are locked. Re-run review against the current kit.

• Insights: citation coverage, most-flagged rules, most-overridden rules, ambiguous-guideline signal, decision rate, average review time.

2. Demo access

Creator	ananya@northwind.example · password demo123
Creator	rohan@northwind.example · password demo123
Reviewer	kabir@northwind.example · password demo123
Admin	meera@northwind.example · password demo123

3. Technical architecture and implementation

Frontend	React 18 + Vite 5, handcrafted CSS; the production build is served by FastAPI from the same origin.
Backend	FastAPI 0.141 on Python 3.14+; SQLModel + SQLite; signed HttpOnly cookie sessions.
Images	Pillow 12 + numpy 2.5: CIELAB ΔE palette checks and WCAG contrast.
Documents	pypdf 6 for PDFs; zip/XML extraction for DOCX.
Review engine	Deterministic in-process engine by default; optional Gemini 2.5 Flash / Ollama adapters, with Citation Lock enforced on every finding.
Deployment	Render Python web service; health check at /api/health.

Key API surface: /api/session, /api/users, /api/bootstrap, /api/kit (GET/PATCH), /api/drafts, /api/drafts/{id}/decisions, /submit, /approve, /request-changes, /rerun, /api/insights, /api/files/{name}.

4. The three improvements we named and delivered

Round 2 committed to three improvements. All three are implemented and observable in the live app.

• 1. Measured evaluation and false-positive reduction — a repeatable synthetic-labelled harness (backend/evaluate.py, 30 cases) computes precision, recall, F1, and false-positive rate; the engine also deduplicates overlapping claim spans so one clause cannot produce two C-01 findings.

• 2. Editable, versioned Brand Kit + Guideline Health — admins edit rules, facts, palette, banned/preferred language, and tolerances in-app; each save bumps the version and appends a history snapshot, closing the read-only gap.

• 3. Failure recovery and reliability — clear 4xx feedback for missing title/caption, unreadable or oversized uploads, empty documents, duplicate decisions, unauthorized actions, and locked approved drafts; signed sessions survive refresh; unit tests cover the engine and decision application.

5. Testing performed

• Engine unit tests (pytest): 7 tests pass — claim overlap dedupe, approved-fact silence, banned promises, urgent tone, PII masking, and stale-quote decision handling.

• Evaluation harness (backend/evaluate.py): 30 synthetic-labelled cases — precision 0.900, recall 0.900, F1 0.900, false-positive rate 0.100.

• API smoke tests: password login for all three roles; kit patch version bump; PII-masked caption through the full review path.

• Browser end-to-end on the deployed link: sign-in page shows credentials on the same page; admin signs in with email + password; edits a rule and saves; version advances 4.3 to 4.4.

• Failure-path checks: zero-finding clean drafts pass; open findings block submit/approve; approved drafts are locked; reviewer cannot create drafts; creator cannot approve; duplicate actions return clear 4xx errors.

• Deployed health check: https://tech-coders.onrender.com/api/health returns 200 with engine status ready.

6. Reliability notes

• All state is server-side; refreshing the page keeps the session and the draft.

• The same action twice returns a clear 400 rather than a crash.

• Empty brand-kit data falls back to the seeded kit.

• Scanned/image-only PDFs are rejected with an explicit “no selectable text” message.

• SQLite is ephemeral on Render unless a persistent disk is attached; the app reseeds the demo workspace on a fresh disk.

7. Known limitations

• Logo clear-space (L-01) is documented in the kit but not yet enforced by a pixel-level check.

• No OCR: scanned PDFs are rejected rather than silently missed.

• PII masking covers emails, phone numbers, and Aadhaar-style IDs; it is not a full redaction product.

• The evaluation harness is synthetic-labelled; the real 30-draft set is the first post-event milestone.

• Mobile layouts are implemented in CSS but were not browser-tested on physical devices.

8. Product roadmap

Next 0–3 months	Recruit pilot brand teams; collect 30 real drafts; publish measured precision/recall and inter-rater agreement.
3–6 months	OCR for scanned documents; pixel-level logo clear-space; exportable audit reports.
6–12 months	Figma/Canva integrations, video/audio review, verified publishing hand-off, sector compliance packs.

9. Team and declarations

Team	A2Z Creators — Venkata Narayana G V (Team Leader, Developer), Mahesh Babu Ch (Research & Analysis).
Build window	All product code was written during the Tech Circuit 2026 event window.
AI disclosure	This document and the code were prepared with AI assistance. The product uses a deterministic local review engine by default; optional external model adapters are disclosed in the UI when enabled.
Privacy	PII is masked on ingest; demo facts are synthetic; no customer content is used for training.